

Statement of Purpose

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My path into offshore renewable energy is recent but deliberate. I am completing an **M.Eng. in Civil Engineering at the South China University of Technology (SCUT)**, and during the first year of my graduate studies my advisor introduced me to the structural analysis of floating offshore wind platforms. I worked across the field's standard benchmark systems, including the OC4 and DeepCwind semi-submersibles paired with the NREL 5 MW reference turbine, the OO-Star and WindFloat concepts, and the DTU 10 MW reference rotor for next-generation studies. That period built my command of the aero-hydro-servo-elastic modelling stack, including **AQWA, OpenFAST, and the F2A framework**, and gave me my first sense of how tightly the disciplines that govern these structures are interwoven.

As I worked through these benchmark systems, another problem emerging in the offshore wind sector began to sharpen the direction of my own research: how to **structurally repurpose the industry's first generation of decommissioned blades**. This became the spine of my master's research, unfolding in three connected stages from material to joint to structure. In the first stage, I designed and executed a mechanical characterisation programme for repurposed blade GFRP, fabricating over one hundred specimens across tensile, compressive, shear, and flexural tests. I derived design values via ANOVA and eigenvalue analysis and proposed position-specific repurposing strategies along the blade. A life-cycle assessment in the same study showed a 71.5% reduction in carbon footprint and a 37.4% reduction in structural mass for a GFRP-based wave energy converter relative to an all-steel equivalent; I led the modelling and analysis for the resulting paper in *Polymers* (SCI Q1).

In the second stage, I designed and tested hybrid bonded-bolted lap joints between repurposed GFRP and duplex stainless steel, identifying the adhesive layer as the governing failure mode with the bolts providing structural redundancy, and validating the behaviour in Abaqus; this first-author manuscript is currently under submission. The third stage, ongoing, takes the problem to the system level: **a one-way coupled hydrodynamic-to-structural analysis of an articulated two-body wave energy converter** built from repurposed GFRP, using **WEC-Sim and Abaqus** linked through custom subroutines I wrote to map the wave loads, with scaled wave-flume testing planned for validation. Alongside this main line, my graduate years have coincided with a period of rapid progress in machine learning, and I took the initiative to build algorithmic competence in parallel. I chose a project adjacent to my structural work, a **deep reinforcement learning approach to operations and maintenance scheduling for offshore wind farms**, precisely because it allowed me to develop end-to-end coding and algorithmic skills on a problem I already understood from the engineering side.. The work was accepted as a first-author paper at the 2026 International Conference on Energy, Power and Electrical Technology (ICEPET 2026), and it gave me the confidence to think of machine learning not as an external tool but as something I could integrate into structural and hydrodynamic research on my own terms.

This direction draws on training that began earlier. As an undergraduate I worked on the

performance of a **novel graded-yielding metallic damper** in steel-frame structures, which gave me a complete cycle of structural analysis from problem definition through finite-element modelling to performance evaluation. A separate undergraduate project on coconut-fibre-reinforced concrete gave me my first exposure to fibre-matrix composite behaviour, a foundation that has carried directly into my current work on GFRP. More than the specific topics, what stayed with me from these early projects was the pull of research itself, of working on a problem long enough to start understanding it from the inside, and that habit has shaped how I approach my graduate work: I have picked up the aero-hydro-servo-elastic modelling stack, the WEC-Sim and Abaqus coupling pipeline, and the algorithmic side of my reinforcement learning project largely on my own, treating each unfamiliar tool as another part of the problem to be learned.

The third stage of my master's research has brought me face to face with the kind of problem I want to spend my doctorate on. Analysing the articulated WEC through a one-way coupled workflow has been clarifying in two ways. First, it has sharpened my sense of where the engineering bottleneck for this class of device sits. Across the WEC literature, the structural body and primary moving structure consistently account for the largest share of capital cost, so the route to commercial viability runs as much through integrated structural design as through component efficiency: how multiple devices respond as arrays rather than as single units, how the choice of power take-off strategy feeds back into structural loading, and how these systems survive the extreme sea states that dominate design loads. Second, wave energy remains largely at the laboratory and pre-commercial stage, while floating offshore wind has matured to the point where established benchmarks exist, even as the rapid scaling of turbine size is reopening structural questions thought to be settled. Both situations make a stronger case for working on these devices as complete engineering systems rather than as isolated structural objects.

The doctoral research I want to pursue follows directly from this. I want to work on the **system-level structural analysis and design of offshore renewable energy devices**, with three connected threads at its core: the extension from single-body to array-level hydrodynamic and structural interaction, the two-way coupling between power take-off control strategy and structural load, and the survivability of these devices, particularly their hinged or articulated connections, under extreme sea conditions. Working on these threads will require moving beyond one-way coupling toward more fully two-way or fluid-structure-interaction approaches where the problem demands it, and the parameter spaces involved are large enough that data-driven methods can earn their place as targeted tools for accelerating parameter sweeps, screening array configurations, and identifying critical load cases, in support of the physics-based analysis rather than as a replacement for it.

Looking ahead, I want to stay in academia and keep working in this area for the long run. As compute and energy demand grow together, offshore renewable energy is becoming part of the infrastructure that supports them, and the engineering problems standing in its way are exactly the system-level structural problems I want to work on.